

Course code: CSE251; Course name: Electronic Circuits

Experiment no.: 1

Title: Verification of Ohm's Law

Objectives:

1. To experimentally verify Ohm's Law

Introduction:

Ohm's law describes mathematically how voltage 'V', current 'I' and resistance 'R' in a circuit are related. According to this law:

"The current in a circuit is directly proportional to the applied voltage and inversely proportional to the circuit resistance".

Formula for voltage:

For a constant value of R, V is directly proportional to I i.e. $V = IR$

Formula for current:

For a constant value of V, I is inversely proportional to R i.e. $I = V/R$

Circuit Diagram:

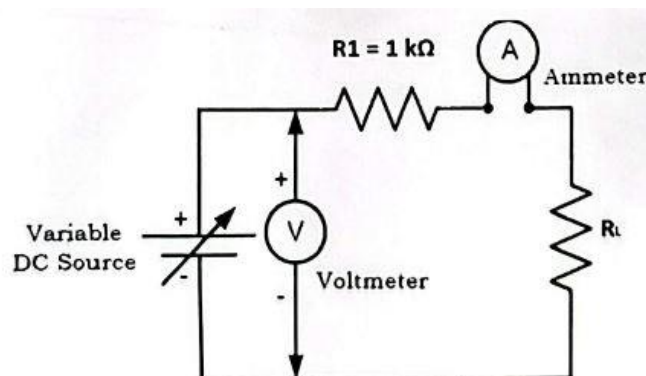


Figure: Circuit set up to verify Ohm's Law.

Equipments and Components Needed:

1. Breadboard
2. DC Variable Power Supply
3. Resistor
4. Connecting Wires
5. Ammeter

Lab Procedure:

- 1) Connect the Power Supply and ammeter in series.

- 2) Now the resistance is adjusted to a fixed value (____ $k\Omega$) so that a constant current flows through the wires.
- 3) Set the initial voltage as ____ V and note down the ammeter reading “I” across the circuit.
- 4) Now increase the initial voltage as ____ V and note down the ammeter reading “I” again .
- 5) The experiment is repeated for different values of voltages up to ____ V and the corresponding current values are noted.
- 6) Calculate the value in each trial. These values will be found to be constants. Thus verifying Ohm's law.

Table 1. Experimental Datasheet.

Trial no.	Voltage Readings	Current Readings